Day 12 Student Notes: Claude for Kubernetes

> **Today in one line:** read what Kubernetes tells you (kubectl output), and let Claude help you **write, review and diagnose** manifests, while you check the evidence.

Kubernetes in plain words

Word	Meaning
Pod	The smallest unit: one running copy of your container
Deployment	Keeps N pods running, replaces failed ones, rolls out updates
Service	One stable address in front of the pods
Ingress	Rules that send outside web traffic to the right Service

```
You: I want 3 copies of shop-api
        |
   Deployment keeps 3 Pods running
        |
   Pod   Pod   Pod
     \\   |   /
   Service = one stable address
```

Four parts of every manifest

```
apiVersion: apps/v1     which API
kind: Deployment        what to create
metadata:               name and labels
  name: shop-api
spec:                   what you want
  replicas: 3
```

Labels connect things: the Service finds pods by their label.

kubectl commands

```
kubectl get pods
kubectl describe pod NAME
kubectl logs NAME
kubectl apply -f file.yaml
kubectl rollout status deployment/NAME
kubectl rollout undo deployment/NAME
kubectl delete -f file.yaml
```

Only look: get, describe, logs. **Change things:** apply, rollout undo, delete. Read before you press Enter.

Pod status words

Status	Meaning	Look at
Running	Container is up (check READY too)	logs if the app misbehaves
Pending	No node can take the pod yet	describe: Events
ImagePullBackOff	Cannot download the image (wrong name or tag)	describe: Events
CrashLoopBackOff	Starts, crashes, restarts again and again	logs, then describe

Debug loop

1. Collect: **kubectl get pods**, **kubectl describe pod NAME**, **kubectl logs NAME**.
2. Ask Claude: what is wrong, which line proves it, what is the fix?
3. **You** check the evidence line in the output.
4. Fix, apply, check again.

Never paste real secrets or company logs into Claude.

Review checklist

Check	Why
Pinned image tag (not :latest)	Same version every time
Requests and limits	No pod can starve the others
Readiness and liveness probes	Traffic only goes to healthy pods
Not root, not privileged	Less damage if the app is hacked
Secrets not in the YAML	Passwords belong in a Secret
More than 1 replica	One pod dying should not mean an outage

The steps and prompts (also in Day12_Prompts_To_Paste.txt)

Step 1: Claude writes the manifests

```
Write a Kubernetes Deployment and a Service for a web app.
Image: nginxinc/nginx-unprivileged:1.27 (it listens on port 8080). 2 replicas. The
Service listens on port 80.
Add resource requests and limits and a readiness probe. Explain each part in simple
words.
```

Check: kind Deployment and Service, replicas 2, port 80 to targetPort 8080, labels match, resources and readinessProbe present.

Step 2: Claude reviews a risky manifest

```
Review manifests/risky-deployment.yaml.
List every risky or missing setting, most serious first, with one line on why.
Then write a fixed copy as manifests/fixed-deployment.yaml. Do not edit the
original.
```

Open manifests/risky-deployment.yaml first and find the risks yourself. There are six.

Step 3: CrashLoopBackOff

```
A pod is failing. Read the files in the case1_crashloop folder.
Tell me in simple words: 1) what is wrong, 2) the evidence line, 3) the fix.
Do not change any file yet.
```

```
Now write the fixed manifest as case1_crashloop/deployment-fixed.yaml.
Change only what is needed and tell me what you changed.
```

Find the evidence line in logs.txt yourself before you trust Claude.

Step 4: two more cases

```
Do the same for the case2_pending folder.
What is wrong, what is the evidence line, and what is the fix?
```

Then do the same for case3_imagepull. Answers: **case 2** asks for 64Gi memory and no node has it (fix: a small request such as 128Mi). **Case 3** uses image tag busybox:1.36.99, which does not exist (fix: busybox:1.36).

Key words

Word	Meaning
Replica	One copy of a pod
Label and selector	A tag on pods, and the rule a Service uses to find them
Request and limit	The minimum memory/CPU a pod asks for, and the maximum it may use
Readiness probe	A check that decides if a pod can receive traffic
Liveness probe	A check that decides if a pod must be restarted
Events	The list at the bottom of kubectl describe, where Kubernetes explains itself
Rollout	Replacing old pods with new ones; rollout undo goes back
Helm	A package manager for Kubernetes; values.yaml holds its settings

Check yourself

1. What is the difference between a Pod and a Deployment?
2. A pod shows Pending. Which command do you run first and what do you read?

3. Name three risky settings in a Deployment.
4. What does CrashLoopBackOff mean, and where do you look?
5. What does a Service do?

Answers

1. A Pod is one running copy. A Deployment keeps the wanted number of pods running.
2. kubectl describe pod NAME, then the Events at the bottom.
3. Any three of: :latest, one replica, no limits, no probes, privileged, password in plain text.
4. The container keeps crashing and restarting. Look at kubectl logs, then describe.
5. It gives the pods one stable address.

Take-home task (30 minutes)

1. Ask Claude for a Helm values.yaml for the Step 1 app (prompt below) and read each key.
2. Optional, only if you have a **throwaway** cluster (Docker Desktop Kubernetes, minikube or kind): run **kubectl apply -f case1_crashloop/deployment.yaml**, watch **kubectl get pods**, then apply your fixed file. Clean up with **kubectl delete -f**.

```
Write a values.yaml for a Helm chart of this app: image, replicas, port and
resources.
Explain each key in one line.
```

Everything in this kit is made up. Never practise on a real company cluster.

> **Next class, Day 13:** Claude for Ansible: automating servers.